UTSAV MURARKA

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WORK EXPERIENCE

Senior Software Engineer | ServiceNow — Mar 2026 – Present

- Tech Lead for AI Response Assist – a GenAI feature that drafts answers to enterprise risk and compliance assessments from past responses and uploaded evidence, cutting completion time by 85%. Leading a 3-engineer pod from design to GA [Feature Demo].
- Architected the RAG pipeline: hybrid multilingual keyword + E5 semantic retrieval, candidate filtering, and a tunable multi-factor ranker that sends the top-3 context to the LLM, supporting 23 languages.
- Built document-grounded answering over PDF, DOCX, XLSX and image evidence through a document intelligence pipeline, with explicit "insufficient evidence" handling to suppress hallucinated answers.
- Designed a fault-tolerant asynchronous response generation system with hierarchical transaction management and tracking, explicit error states, and partial-completion handling with granular retries. Achieved a p95 latency of ~4 seconds per question.
- Developed an LLM-as-a-judge evaluation suite for metrics like correctness, completeness and faithfulness. Tuned prompts to achieve scores above 90% on a 1000-question golden dataset across 3 LLM providers.
- Defined API contracts and a plugin interface for the AI pipeline, letting 4 consuming product teams integrate their own data sources in parallel without changes to the core engine.
- Drove code quality for the team as primary reviewer, with 500+ pull requests reviewed across 25+ repositories, enforced strict linting rules, pre-commit hooks, Jenkins CI health-checks, and release-readiness checks like localization and security.
- Drove AI-DLC (AI-driven development lifecycle) for the team: rolled out an autonomous coding agent that turns GitHub issues into pull requests and a reviewer agent that reviewed 800+ PRs across 6 repositories. Reduced defect turnaround time by 40%.

Software Engineer | ServiceNow — Mar 2024 – Feb 2026

- Owned and delivered Smart Assessment Engine's collaboration capability, enabling multi-contributor support with real-time synchronization and presence indicators for seamless concurrent authoring; used in 40% of all vendor risk assessments [Docs].
- Designed and delivered a no-code Condition Builder and evaluation framework that powers key features for Smart Assessment Engine like conditional visibility, preferred answers, conditional evidence attachments – enabling a successful GA launch.
- Architected and developed a low-code predicate-builder framework for the Post Assessment Actions Rule Engine, automating actions like emails and approvals, enabling workflow automations linked to Smart Assessments [Docs].
- Partnered with PM, UX and technical writers to define the roadmap for the Smart Assessment Engine used by 20+ product applications. Led the platform-side integration to onboard Field Service Management as a new consumer [Integration Demo].

Associate Software Engineer | ServiceNow — Jul 2022 – Feb 2024

- Delivered autosave for Risk Assessments: UI, REST APIs, WebSocket refresh, reducing clicks per assessment by 60% (~200 to ~80).
- Built a user-role hierarchy validation tool saving 20+ QA hours per release, and a role-based-access licensing simulation tool.
- Achieved WCAG 2.0 compliance by fixing 15+ issues for compatibility with screen readers, zoom, and high contrast modes.

Software Engineer Intern | Amazon — Jan 2022 – Jun 2022

- Productionized a proof-of-concept of the selection safe-listing feature in Amazon's internal GraphQL service.
- Authored the low-level design and implemented it in Java with shared utilities reused by the persisted-queries module.

Software Engineer Intern | ServiceNow — May 2021 – Jul 2021

- Designed and developed a proof-of-concept for a feature to visualize and manage hierarchical entity relations in a database.
- Created a drag-and-drop UI using D3.js library, and REST APIs to fetch the hierarchical data by performing recursive queries.

Summer Intern | Infibeam Avenues — May 2019 – Jul 2019

- Implemented an AI Chatbot application using Django, PostgreSQL and Dialogflow API to conduct interviews.

EDUCATION

Birla Institute of Technology and Science, Pilani – K. K. Birla Goa Campus — Aug 2017 – Jun 2022
B.E. Computer Science & M.Sc. Physics (Integrated Dual Degree) — **CGPA: 8.90 / 10**

SKILLS

Languages: JavaScript, Java, C++, Python, SQL, GraphQL, HTML, CSS
Frameworks: REST APIs, WebSockets/Pub-Sub, React, Node.js, Lit, Web Components, PostgreSQL, ServiceNow
AI: LangChain, LangGraph, RAG, Embeddings, Hybrid/semantic search, Prompt engineering, LLM-as-a-judge, Agentic Harnesses

RESEARCH

The Neutron Star Outer Crust Equation of State: A Machine Learning Approach | Research Project at BITS [Publication] — Jan 2022

- Lead author of the publication in the Journal of Cosmology and Astroparticle Physics, January 2022 Edition.